

ONLINE RETAIL SALES DATABASE MANAGEMENT SYSTEM

Introduction

The Online Retail Sales Database Management System is a MySQL-based database project developed to manage and analyze the operations of an online retail business. The system stores information about customers, products, categories, orders, order details, and payments in a structured database. It helps reduce data duplication, maintain data accuracy, and retrieve useful business information through SQL queries. The project also demonstrates database design, relationships, constraints, joins, views, subqueries, Common Table Expressions, and window functions.

Abstract

This project focuses on designing and developing a relational database system for an online retail business. The database is created using MySQL and contains six main tables: Categories, Customers, Products, Orders, Order_Details, and Payments. Primary keys and foreign keys are used to establish relationships between tables and maintain data integrity. Sample retail data is inserted into the database for testing and analysis. SQL queries are used to generate customer-wise sales, product performance, category-wise sales, monthly sales, payment analysis, and order-status reports. Advanced SQL techniques such as CTEs, subqueries, ranking functions, and window functions are also implemented. The system provides a structured and efficient method for managing retail sales information.

Problem Statement

Online retail businesses generate large amounts of customer, product, order, and payment information. Managing this information manually can result in duplicate records, incorrect data, difficulty in searching information, and poor reporting. A centralized database system is required to store and manage the information efficiently. This project solves the problem by creating a relational database that organizes retail information into connected tables and provides SQL-based analysis for business reporting.

Objectives

- To design a structured database for an online retail business.
- To store customer, product, order, and payment information efficiently.

- To establish relationships between different database tables.
- To maintain data accuracy using primary keys, foreign keys, constraints, and validation rules.
- To reduce unnecessary data duplication through database normalization.
- To perform SQL queries for retrieving important business information.
- To analyze customer purchasing behavior and sales performance.
- To identify top-selling products and profitable product categories.
- To generate monthly sales and order reports.
- To demonstrate advanced SQL concepts such as CTEs, subqueries, ranking, and window functions.

Tools Used

- MySQL
- MySQL Workbench
- SQL
- dbdiagram.io
- Microsoft Excel for preparing sample data

Database Design

The database is named `online_retail_db`. It contains six major tables: Categories, Customers, Products, Orders, Order_Details, and Payments. Each table is designed for a specific purpose and connected using primary keys and foreign keys.

The Categories table stores product category information. The Customers table stores customer details such as name, email, phone number, city, state, and registration date.

The Products table stores product names, prices, stock quantities, suppliers, and category information.

The Orders table stores customer orders, order dates, order status, and total amounts.

The Order_Details table records the products included in each order along with their quantities and prices. The Payments table stores payment dates, payment methods, payment status, and payment amounts.

Relationships

The Categories table is connected to the Products table because each product belongs to a category.

The Customers table is connected to the Orders table because a customer can place multiple orders.

The Orders table is connected to the Order_Details table because one order can contain multiple products.

The Products table is connected to the Order_Details table because a product can appear in multiple orders.

The Orders table is connected to the Payments table because payment information is associated with customer orders.

Normalization

The database follows the principles of normalization to improve data organization and reduce duplication.

First Normal Form ensures that each table contains atomic values and that repeating groups are avoided.

Second Normal Form ensures that non-key attributes depend on the complete primary key of the table.

Third Normal Form ensures that non-key attributes do not depend on other non-key attributes.

Normalization improves data consistency and makes the database easier to maintain.

SQL Operations

Different SQL operations are implemented in the project to manage and analyze the retail data. DDL commands are used to create the database and tables. Constraints are applied to maintain data integrity. DML commands are used to insert sample records.

SELECT queries are used to retrieve required information. JOIN operations combine information from multiple tables. GROUP BY and aggregate functions such as COUNT, SUM, and AVG are used for business analysis. WHERE and HAVING clauses are used to filter the required records.

Sales Analysis

The database can be used to analyze total sales generated by different products and categories. Customer-wise purchase analysis helps identify customers who contribute significantly to sales. Product-wise analysis helps identify products with high sales quantities.

Category-wise analysis provides information about the performance of different product categories. Monthly sales analysis helps understand sales trends over time. Payment analysis provides information about different payment methods and their statuses.

Views

SQL views are created to simplify frequently required reports. The project includes views such as Sales Report, Customer Sales Report, Product Sales Report, Category Sales Report, and Monthly Sales Report.

These views allow users to retrieve important information without repeatedly writing complex JOIN and aggregation queries.

Advanced SQL

Advanced SQL techniques are used to improve the analytical capabilities of the project. Common Table Expressions are used to create temporary result sets for complex queries.

Subqueries are used to compare individual records with calculated values such as average sales. Ranking functions are used to rank customers, products, and categories according to their sales performance.

Window functions are used to calculate running totals and customer-specific rankings. These techniques demonstrate the analytical capabilities of SQL beyond basic database operations.

Results

The developed database successfully stores and manages online retail information. Relationships between tables work through primary and foreign keys. SQL queries successfully generate customer, product, category, order, payment, and monthly sales information.

The project also demonstrates advanced SQL analysis using CTEs, subqueries, ranking functions, and window functions. The created views provide simplified access to important business reports.

Conclusion

The Online Retail Sales Database Management System provides a structured solution for managing online retail data. The project demonstrates important database concepts

including database design, normalization, relationships, constraints, SQL queries, JOINS, views, and advanced SQL analysis.

The system can help a retail business organize its information and generate useful reports for understanding customers, products, orders, payments, and sales performance. The project also provides a strong practical demonstration of MySQL and SQL skills for an MSc Data Science student.